



AC CIRCUITS



Introduction:

Alternating current is the back bone of electrical engineering and physics. In addition, we will explore the dynamic nature of AC current, which is quite different when comparing with its counterpart, Direct Current (DC).

Cycle:

One complete set of both positive and negative values of an alternating quantity is known as a cycle.

Time Period (T):

The time period of an alternating current (AC) or voltage wave is the duration it takes to complete one full cycle.

Frequency (f):

It refers to the number of complete cycles of the current waveform that occur in one second.

Instantaneous Peak Value:

The instantaneous peak value refers the maximum amplitude of the alternating current or voltage at any specific point in time.

For a sine wave, this is the highest positive or negative value reached during a cycle. It is usually denoted as I_{peak} for current or V_{peak} for voltage.

Root Mean Square (rms) Value:

The average value of voltage or current is not used in electric power calculations. The reason being that the AC cycle consists of positive and negative half cycles so the average value is zero.

However, there exists a mathematical relation between the peak value V_o of alternating voltage and a direct current (d.c) voltage that yields an equivalent average electrical power. This constant d.c. voltage is referred to as the root-mean-square (r.m.s)

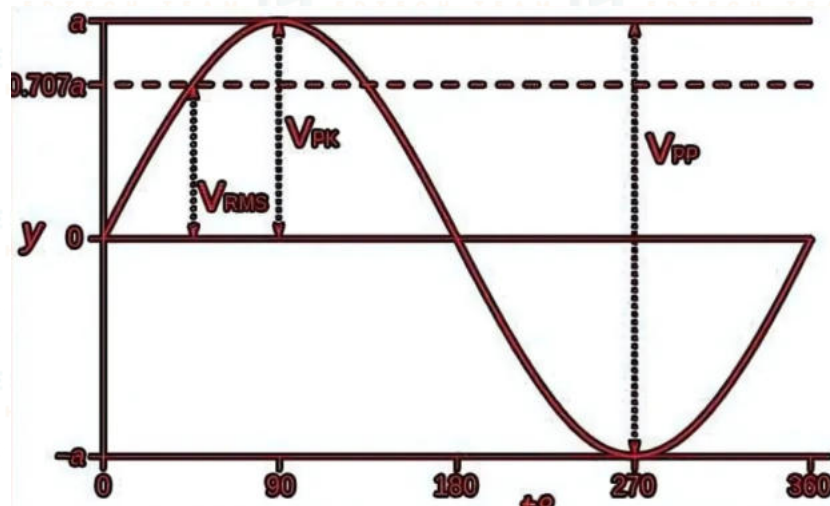
value of the alternating voltage V_{rms} . The d.c voltage is approximately 70% of V_{peak} as shown in figure.

For a sinusoidal wave, the rms value is approximately 0.707 times the peak value.

Mathematically,

$$V_{\text{rms}} = 0.707 \times V_{\text{peak}} \text{ for voltage}$$

$$\text{and } I_{\text{rms}} = 0.707 \times I_{\text{peak}} \text{ for current.}$$



Alternating Current OR voltage equation:

Alternating Current (AC) or Voltage refers the type of electrical flow or potential difference that periodically changes direction and magnitude over time.

The fundamental equation describing AC voltage or current in a sinusoidal waveform is:

$$V = V_{\max} \sin(\omega t)$$

$$I = I_{\max} \sin(\omega t)$$

Where,

V, I represent the instantaneous value of the voltage and alternating current respectively

$V_{\max}$ is the amplitude of the waveform, which represents the maximum value of alternating voltage and alternating current respectively

ω is the angular frequency, given by $\omega = 2\pi f$

Phase of AC:

The phase difference between two waves carries more important than their magnitudes. The phase describes the relative positions of these waveforms.

Hence in alternating current (A.C.) circuits, the concept of phase refers to the relationship between different waveforms as time passes.

Now we discuss different states of phase within AC circuit.

In-phase:

When two waveforms have the same frequency and reach their peak values or zero values at the same time, they are said to be in phase.

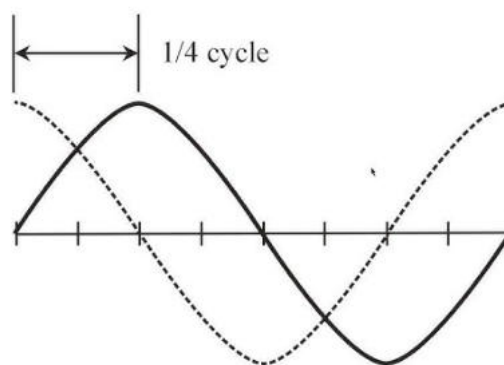
Phase Lag:

If one waveform reaches its peak or zero value after the other, there is a phase difference, and the second waveform is said to lag the first.

Phase Lead:

Conversely, if one waveform reaches its peak or zero value before the other, the second waveform is said to lead the first.

The representation of phase lead and lag between two alternating quantities is effectively demonstrated by showing the two AC quantities as vectors also known as "Phasor".



AC through a resistor:

Consider a circuit comprising a pure resistor (R) connected to an alternating voltage source, as depicted.

The alternating voltage induces oscillatory motion of free electrons within the resistor, constituting the alternating current. At any given time ' t ', the potential difference across the resistor's terminals is expressed

$$V = V_{\max} \sin(\omega t)$$

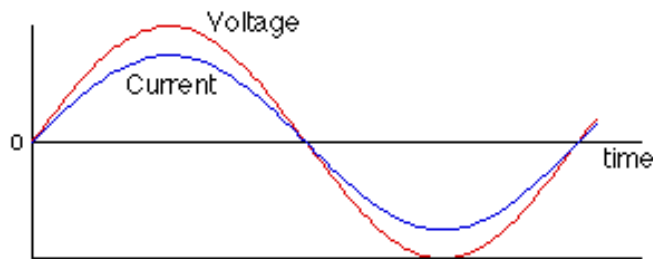
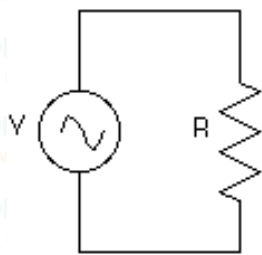
where V_0 signifies the peak value of the alternating voltage. The circuit's current (I) is governed by Ohm's law:

$$I = V_0 / R \sin(\omega t)$$

Or

$$I = I_{\max} \sin(\omega t)$$

The current achieves its maximum value when $\sin(\omega t) = 1$.



Phasor diagram of Resistive circuit:

The current I flowing through a resistor R is in phase with the voltage V_x across the resistor.

This alignment can be visually represented on a phasor diagram by depicting a vector (I_a) that coincides with the voltage (V_s).

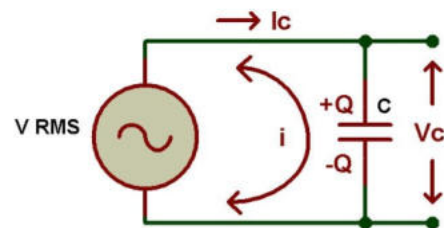
The power loss in a resistor in an AC circuit is a result of the conversion of electrical energy into heat due to the resistance of the resistor. The instantaneous power in the resistance is can be expressed using the formula

$$P = I^2 R$$

$$P = V_{rms} I_{rms}$$

A.C through capacitors:

Now consider a capacitor connected to the AC source as shown in figure. During the positive half cycle of alternating voltage, the electrons flow from upper capacitor plate to the source leaving it as positively charged $+Q$ and source supplies equal number of electrons to the lower plate making it as negatively charged $-Q$ as shown in figure 20.9.



During the negative half cycle of alternating voltage, the direction of motion of electrons is also reverted resulting in the capacitor plates becoming charged in the opposite manner and current flows in opposite direction, hence capacitor charges and discharges. In this way the alternating current flows through the capacitor.

If alternating source voltage $V = V_{max} \sin(\omega t)$ is applied to the capacitor the charge on any plate of capacitor is given by

$$q = CV = CV_{max} \sin(\omega t) \text{ ---- (1)}$$

From above equation it is clear that q and V are in phase. Due to the applied current flows through the capacitor as given by

$$I = dq/dt \text{ ---- (2)}$$

Substitute equation (1) in (2), we get

$$I = C(dV/dt)$$

$$I = C\omega V_{\max} \sin(\omega t)$$

Or

$$I = C\omega V_{\max} \cos(\omega t)$$

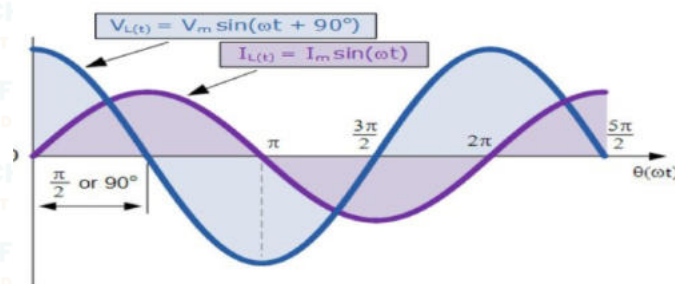
Multiply and divide right side of equation (3) by ω :

$$I = V_o \sin(\omega t / X_c)$$

$$I = I_{sx} \sin(\omega t + \pi/2) \text{ ---- (4)}$$

Equation (4) shows that in pure capacitive circuit the current exhibits sinusoidal variation and it leads the voltage by 90 degrees.

$$I = I_{sx} \sin(\omega t + \pi/2) \text{ ---- (4)}$$



Reactance of Capacitor:

In a purely capacitive circuit, capacitive reactance represents the opposition to alternating current flow. Similar to resistance, reactance is measured in Ohms, but it is denoted by the symbol **X_c** to differentiate it from purely resistive values.

Since,

$$I = V_s / X_c$$

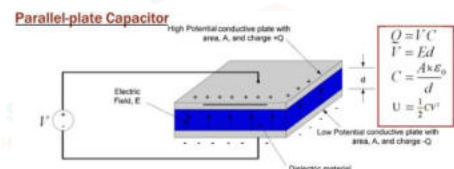
Similar to ohm's law, the term in denominator of equation represents the opposition to flow of current; in this case it is called reactance of capacitor.

Hence,

$$X_c = 1 / \omega C$$

Since,

$$X_c = 1 / 2\pi f C$$



The capacitive reactance shows an inverse relationship with the frequency of the applied alternating voltage.

Consequently, for lower frequencies, the reactance of capacitor increases, while at high frequencies, the reactance decreases.

A.C through inductor:

An inductor is a passive electrical component which can be formed by winding a conductor wire over an insulating object, such as pencil. The primary purpose of an

Inductor is to oppose changes in current.

It resists the flow of alternating current (AC) and stores energy in its magnetic field during the on time of the AC cycle and releases it during the off-time. Solenoid is an example of inductor.

Consider an inductor in form of solenoid connected with A.C source.

When the switch is closed current start to flow through the inductor, notice that the magnitude and direction of current is changing hence associated magnetic field also varies due to which an induced emf is set up in the coil to oppose the change in accordance with the Lenz law. The magnitude of induced emf is given by

$$\varepsilon = L (di/dt)$$

Therefore, to sustain the current, the applied voltage must match the back electromotive force (EMF). Thus, the magnitude of voltage supplied to the coil is expressed as

$$V = L (di/dt)$$

The alternating voltage produces a sinusoidal current given by

$$I = I_{sx} \sin(\omega t)$$

Hence above equation becomes

$$V = L (I_{sx} \sin(\omega t)) / dt$$

or

$$V = \omega L_o \text{ Cos} \omega t$$

Where $\omega L_o = V_o$

$$V = V_o \text{ Cos} \omega t$$

Using that $\cos \omega t = \sin(\omega t + \pi/2)$

$$v = V_o \sin(\omega t + \pi/2)$$

Choke coil:

A choke coil, commonly known as choke, is an inductor used in electronic circuits. It is used to block high frequencies above a certain frequency, while allowing the direct current to pass through.

Inductive Reactance:

The term inductive reactance refers to the opposition that an inductor offers to the flow of alternating current. It is denoted by X_L . It is measured in ohms. The inductive reactance can be calculated by following the method.

The formula for inductive reactance is given by:

$$X_L = \omega L$$

$$\text{Or } X_L = 2\pi fL$$

where:

X_L is the inductive reactance,

f is the frequency of the AC signal,

L is the inductance of the inductor.



RLC Circuits:

An RLC circuit is an electrical circuit consisting of a resistor (R), inductor (L), and capacitor (C), connected in series or parallel.

RLC circuits are fundamental in electronics for filtering signals, tuning circuits, and resonant applications due to their ability to manipulate frequencies and impedance.

Impedance triangle:

A circuit containing resistor, inductor and capacitor offer opposition to flow of current due to all these circuit elements known as impedance. It is denoted by Z. The impedance triangle:

$$Z = \sqrt{R^2 + X_L^2}$$

Where

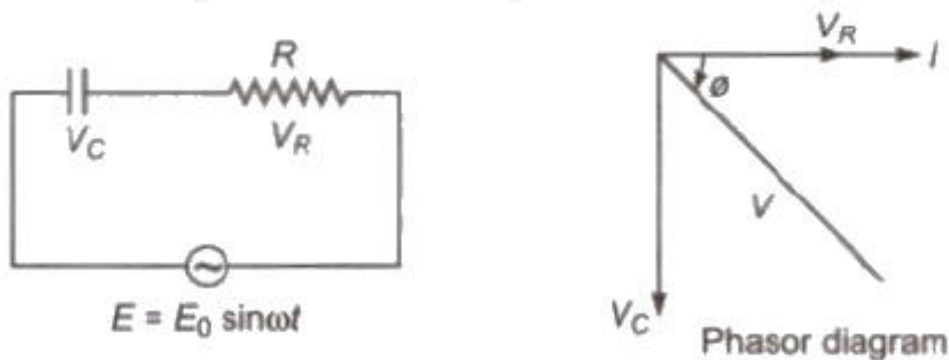
Z = impedance

R = Resistance in circuit

X_L² = sum of capacitive

AC through RC Series circuits:

Consider a circuit containing resistor and capacitor connected in series with an alternating voltage source as shown in figure 20.16. Due to series circuit same current will flow through each circuit element. While voltage across resistor will be V_r and V_x across capacitor.



The phasor diagram is drawn taking current as reference direction.

From the phasor diagram, using Pythagoras theorem:

Substituting voltage across resistor V_r and across capacitor V_x in above equation

$$V = \sqrt{(V_r^2 + V_x^2)}$$

$$V = \sqrt{[(I_s R)^2 + (I_s X_x)^2]}$$

$$V = I_s \sqrt{(R^2 + X_x^2)}$$

Comparing equation $V = I_s \sqrt{(R^2 + X_x^2)}$ and $Z = \sqrt{(R^2 + X_x^2)}$, we find that quantity V represents the impedance of Z series circuit.

Phase angle between current and voltage as shown in figure 20.19 is calculated by:

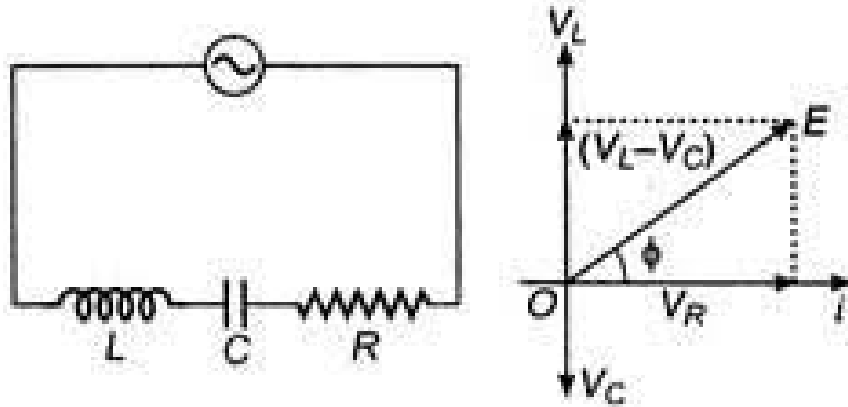
$$\tan(\varphi) = X_x / R$$

or

$$\varphi = \tan^{-1} (X_x / R)$$

AC through RL Series circuits:

Next in order to calculate the impedance of RL series circuit, consider a circuit containing inductor and resistor connected in series with an alternating voltage as shown in figure. The voltage across resistor and inductor in V_R and V_L respectively.



When an inductor is connected with alternating voltage then due to generation of back EMF current lag the voltage by 90° , it is represented by line perpendicular to the reference phasor i.e. current as shown in figure. Also as discussed in section 20.31 in resistor the current and voltage are in phase. In phasor diagram it is V line.

To calculate the impedance of RL circuit, apply Pythagoras theorem on phasor:

$$Z = \sqrt{(R^2 + L^2)}$$

and $Z = \sqrt{(R^2 + L^2)}$ represents the impedance of the RL circuit.

Phase angle between current and voltage as shown in figure is calculated by Figure 20.20

$$\tan(\phi) = X_L / R$$

or

$$\phi = \tan^{-1} (X_L / R)$$

RLC Series AC Circuit:

An RLC series circuit consists of a resistor (R), inductor (L), and capacitor (C) connected in series to an AC voltage source as shown. In such circuit, the electrical components influence the behavior of the current flowing through the circuit.

Power in A.C Circuits:

Power dissipation in AC circuit is expressed as follows:

$$P = VI \cos \phi$$

Where $\cos \phi$ represents power factor.

Resonant Frequency:

For a certain frequency the capacitive and inductive reactance becomes equal, $X_L = X_C$. This frequency is called resonant frequency f_0 and circuit is said to be in resonance state. To calculate the resonant frequency, use the fact that

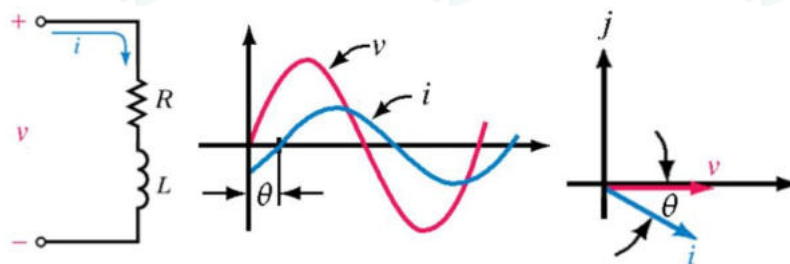
$$\begin{aligned}X_L &= X_C \\1 / 2\pi f_0 &= \sqrt{LC} \\ \text{or} \\ f_0 &= 1 / 2\pi \sqrt{LC}\end{aligned}$$

Parallel RL Circuit:

In some respects, the circuit of figure is similar to the purely inductive parallel circuit.

For instance, applied voltage V is still the quantity which is common to both components and is therefore plotted in standard position in the phasor diagram. Also the magnitude of the individual branch currents is determined by the opposition (reactance) of the individual branches. The figure shows a composite diagram of waveforms and phases.

The phasor diagram shows that the two branch currents are not in phase, it will be necessary to use phasor addition in order to determine the total current.

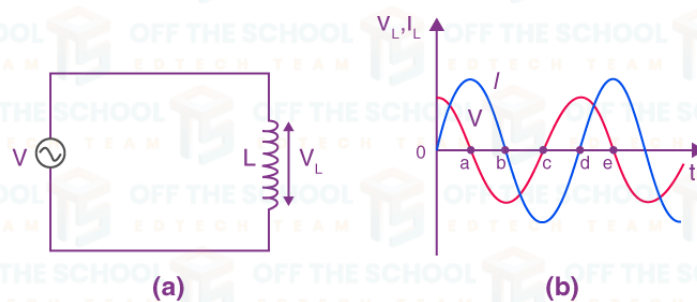


Parallel RC circuits

Parallel RC circuits may be resolved in much the same way as parallel RL circuits.

The figure illustrates a parallel RC circuit.

The figure shows a composite diagram of waveforms and phasors as per circuit conditions. The current phasors I_R and I_C are out of phase; therefore, phasor addition must be used to determine total current. The solving of an RC circuit follows the method previously applied to LR circuits.

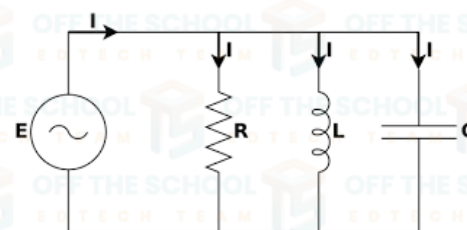


Parallel RLC AC Circuit:

A Parallel RLC AC Circuit one where the resistor, inductor, and capacitor are connected in parallel to each other and the AC source. In this configuration, the voltage across each component is the same, but the currents

through them differ. The parallel arrangement affects the overall impedance and current distribution in the circuit, making it distinct from its series counterpart.

In a parallel circuit, the voltage **V (RMS)** across each of the three elements remains the same. Hence, for convenience, the voltage may be taken as a reference phasor.



In the parallel RLC circuit, the supply voltage **Vs** is common to all three components, while the supply current is consisting of three parts: the current through the resistor (**Ir**), the current through the inductor (**IL**), and the current through the capacitor (**IC**). The current flowing through each branch, and therefore through each component, will differ from one another and from the supply current.

The total current drawn from the supply is not simply the arithmetic sum of the three individual branch currents but their vector sum.

Similar to the series RLC circuit, we can solve this circuit using the phasor or vector method. However, in this case, the vector diagram will use the voltage as its reference, with the three current vectors plotted relative to this reference voltage.

The phasor diagram for an AC RLC parallel circuit is created by combining the individual phasors for each component and adding them vectorially.

Since the voltage across the circuit is common to all three circuit elements, we can use it as the reference vector, with the three current vectors drawn relative to this reference at their corresponding angles. The resulting vector current **I** is obtained by first adding the vectors **IL** and **IC** and then adding this sum to the vector **IR**. The angle between **V** and **I** is

Resonance of Parallel RLC AC Circuit:

A parallel circuit containing a resistance **R**, an inductance **L**, and a capacitance **C**, will produce a parallel resonance (also called anti-resonance) circuit when the resultant current through the parallel combination is in phase with the supply voltage. At resonance there will be a large circulating current between the inductor and the capacitor due to the energy of the oscillations, then parallel circuits produce current resonance.

A parallel resonant circuit stores the circuit energy in the magnetic field of the inductor and the electric field of the capacitor. This energy is constantly being transferred back and forth between the inductor and the capacitor which results zero current and energy being drawn from the supply.

In the solution of AC parallel resonance circuits, we know that the supply voltage is common for all branches, so this can be taken as our reference vector. Each parallel branch must be treated separately as with series circuits so that the total supply current taken by the parallel circuit is the vector addition of the individual branch currents.

$$I_s = V / R$$

$$I_L = V / X_L$$

$$I_C = V / X_C$$

Therefore, **I = vector sum of (I_L + I_C)**

$$I_s = \sqrt{(I_L^2 + I_C^2)}$$

At resonance, currents **I_L** and **I_C** are equal and cancelling giving a net reactive current equal to zero. Then at resonance the above equation becomes.

$$I_s = \sqrt{(I_r^2 + I_c^2)}$$

$$I_r = IR$$

Therefore, the circuit current at this frequency will be at its minimum value of V/R and the graph of current against frequency for a parallel resonance circuit is shown in figure.

The frequency response curve of a parallel resonance circuit shows that the magnitude of the current is function of frequency and plotting this onto a graph shows us that the response starts at its maximum value, reaches its minimum value at the resonance frequency when $I_{r.m.s} = I_r$ and then increases again to maximum as f becomes infinite.

The result of this is that the magnitude of the current flowing through the inductor, L and the capacitor, C tank circuit can become many times larger than the supply current, even at resonance but as they are equal and at opposition (**180° out-of-phase**) they effectively cancel each other out.

As a parallel resonance circuit only functions on **resonant frequency**, this type of circuit is also known as an **Rejecter Circuit** because at resonance, the impedance of the circuit is at its maximum thereby suppressing or rejecting the current whose frequency is equal to its resonant frequency. The effect of resonance in a parallel circuit is also called "current resonance".

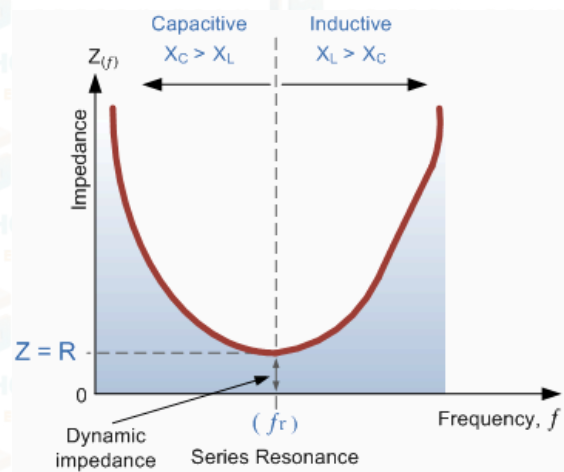
Maximum Power Transfer Theorem:

According to this theorem, maximum power is transferred from a source to a load when the impedance of the source matches with the impedance of the load.

Mathematically, suppose R_s denotes the resistance of load and R_i is source resistance.

Metal Detectors:

The oscillator circuit is used in metal detectors. Metal detectors used for security checks operate on the principle of electromagnetic induction. A simplified explanation of working of metal detector is described below:



1. Generating an Electromagnetic Field:

The metal detector contains a coil of wire through which an electric current flows, creating a magnetic field around the coil. This coil is often housed in a special arrangement, such as a loop or wand.

2. Interaction with Metals:

When a conductive metal object is brought into the vicinity of the electromagnetic field, it disturbs the field. This disturbance induces a secondary magnetic field in the metal object.

3. Eddy Currents:

The changing magnetic field induces circulating electric currents within the metal object, known as eddy currents. These eddy currents, in turn, generate their own magnetic fields.

4. Detection of Changes:

The metal detector has a receiver coil or coils that are in close proximity to the transmitter coil. The receiver coil(s) detect changes in the magnetic field caused by the presence of the metal.

5. Alert Mechanism:

When the metal detector senses a significant change in the magnetic field, indicating the presence of a metal object, it triggers an alert. This alert can be in the form of an audible sound, a visual signal, or both, depending on the design of the metal detector.

The Electrocardiograph:

The electrocardiograph (ECG) is a medical diagnostic tool used to record the electrical activity of the heart over a period of time.

Oscillator Circuit:

An oscillator circuit is an electronic circuit that generates a continuous periodic signal at a specific frequency.

Resonance in Tuning circuit of Radio:

In radio tuning circuits resonance is an important phenomenon that enables the selective reception of a desired radio frequency while rejecting others.

Importance of Broadcasting:

Broadcasting remains a powerful medium with far-reaching impacts on society. It informs, educates, entertains, and connects people, playing a crucial role in cultural preservation, economic development, and social cohesion. Its ability to reach a wide audience makes it an indispensable tool for communication in the modern world.